

Christine P. Lee | Curriculum Vitae, October 2026

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Research Interests

My research in human-computer interaction, AI, and robotics examines how people can retain control over the decisions AI agents make on their behalf. I design and evaluate systems that make people's goals, constraints, and preferences explicit and editable, helping them specify what should guide an agent, verify its behavior, and revise that guidance over time. My work combines empirical user studies, LLM systems, and formal verification, with applications in AI planning, personalization, and household robotics. My future research focuses on agent memory, delegation across multiple agents and people, and embodied agents that infer intent from observation.

Education

- 2022– PHD, Computer Sciences, University of Wisconsin–Madison, expected Spring 2027
Advisor: Dr. Bilge Mutlu. *Doctoral Minor:* Communication Arts
Committee: Dr. Bilge Mutlu, Dr. Min Kyung Lee, Dr. Jodi Forlizzi, Dr. Aws Albarghouthi, Dr. Sharon Li
- 2020–2022 MS, Computer Sciences, University of Wisconsin–Madison
Advisor: Dr. Bilge Mutlu
- 2015–2019 BS, Computer Sciences, *Summa Cum Laude*, Sejong University, Seoul, South Korea

Research Experience

- 2020– *Research Assistant*, University of Wisconsin–Madison, Madison, WI
Mentor: Dr. Bilge Mutlu
- 2026 *Robotics Applied Scientist Co-op*, Amazon Personal Robotics Group (Fauna Robotics), Sunnyvale, CA
Project: Character-based memory architecture for household robots
Mentors: Dr. Hae Won Park, Hani Abdo, and Xiaoxu Liu
- 2020 *Research Intern*, Seoul National University Hospital Biomedical Research Institute, Seoul, South Korea, January to July 2020
Mentor: Dr. Taehoon Ko
- 2019–2020 *Undergraduate Researcher*, CVPR Lab, Sejong University, Seoul, South Korea
Advisor: Dr. Hyun-Joon Moon

Honors & Awards

FELLOWSHIPS, AWARDS & DISTINCTIONS

- 2026 Selected Participant, Future Leaders of AI Doctoral Consortium, ACM AI Leadership Summit 2026 (53 selected out of 142)
Position paper: “Designing Responsible AI Workflows: Preserving Control and Supporting Alignment”
Special Recognition for Outstanding Reviews, ACM CHI Conference on Human Factors in Computing Systems (CHI 2026)
- 2025 Cisco Distinguished Graduate Fellowship, one year of full PhD funding

- 2025– NSF Research Traineeship Program “INTEGRATE” Fellowship, Spring 2025 to present
 Special Recognition for Outstanding Reviews, ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)
 Special Recognition for Outstanding Reviews, ACM Symposium on User Interface Software and Technology (UIST 2025)
- 2024 Doctoral Consortium, ACM CHI Conference on Human Factors in Computing Systems (CHI 2024), with \$1,700 award
- 2021 Summer Research Assistant Award, UW–Madison Computer Sciences Department
 Chosen as one of 15 students in the department based on faculty nomination

PAPER AWARDS

- 2025 Honorable Mention (Top 251 in 5020), ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)
 Kim, Sato, White, Ho, **Lee**, Hwang, & Mutlu: Bridging Generations using AI-Supported Co-Creative Activities
- 2022 Honorable Mention (Top 131 in 2597), ACM CHI Conference on Human Factors in Computing Systems (CHI 2022)
Lee, Cagiltay, & Mutlu: The Unboxing Experience: Exploration and Design of Initial Interactions Between Children and Social Robots

Publications

* indicates equal contribution. **[Award]** indicates a paper award (see Paper Awards).

IN PRESS (1)

Lee, C. P., Lee, M. K.*, & Mutlu, B.* From Tool to Agent: How Worker Needs Reorganize Across the Agentic Roles of Workplace AI. *Proceedings of the 14th International Conference on Human-Agent Interaction (HAI '26)*.

JOURNAL ARTICLES (2)

- 2026 Hwang, Y., **Lee, C. P.**, Porfirio, D., Hiatt, L. M., & Mutlu, B. Formal Methods in Robot End-User Development: Progress, Gaps, and Opportunities. *Foundations and Trends in Robotics*, 13(3), 273–349.
- 2020 Kim, H. K., Choi, S. W., Bae, Y. S., Choi, J., Kwon, H., **Lee, C. P.**, Lee, H.-Y., & Ko, T. [MARIE: A Context-Aware Term Mapping with String Matching and Embedding Vectors](#). *Applied Sciences*, 10(21), 7831.

FULL-LENGTH CONFERENCE PAPERS (8, PLUS 2 UNDER REVIEW)

- Lee, C. P.**, Porfirio, D., Pan, E., & Mutlu, B. PromptPath: Bringing Persistent User Context into LLM Prompt Formulation. *Under review*.
- Lee, C. P.**, Sullivan, D., Wang, X. J., & Mutlu, B. A Communication Initialization Profile (CIP) for Cold-Start LLM Personalization. *Under review*.
- 2026 Wang, X. J., **Lee, C. P.**, & Mutlu, B. [LearnMate²: Design and Evaluation of an LLM-powered Personalized and Adaptive Support System for Online Learning](#). *Proceedings of the 2026 Designing Interactive Systems Conference (DIS '26)*.
- Kim, Y., **Lee, C. P.**, & Mutlu, B. [Speaking with Screens: Design Space and Guidelines for Informational Robot Screens](#). *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI '26)*.
- 2025 **Lee, C. P.**, Porfirio, D., Wang, X. J., Zhao, K., & Mutlu, B. [VeriPlan: Integrating Formal Verification and LLMs into End-User Planning](#). *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*.
- [Award]** Kim, C. Y., Sato, A. J., White, N. T., Ho, H.-R., **Lee, C. P.**, Hwang, Y., & Mutlu, B. [Bridging Generations using AI-Supported Co-Creative Activities](#). *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*.
- 2024 **Lee, C. P.**, Lee, M. K.*, & Mutlu, B.* [The AI-DEC: A Card-based Design Method for User-centered AI Explanations](#). *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*.

Lee, C. P., Praveena, P., & Mutlu, B. [REX: Designing User-centered Repair and Explanations to Address Robot Failures](#). *Proceedings of the 2024 ACM Designing Interactive Systems Conference (DIS '24)*.

Kim, C. Y.*, **Lee, C. P.***, & Mutlu, B. [Understanding Large-Language Model \(LLM\)-powered Human-Robot Interaction](#). *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction (HRI '24)*.

2022 **[Award]** **Lee, C. P.**, Cagiltay, B., & Mutlu, B. [The Unboxing Experience: Exploration and Design of Initial Interactions Between Children and Social Robots](#). *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*.

SHORT PAPERS & EXTENDED ABSTRACTS (4, PLUS 1 UNDER REVIEW)

Lee, C. P., Liu, X., Abdo, H., & Park, H. W. Character-Based Memory: Shaping How Household Robots Interpret Experience. *Under review, ACM/IEEE International Conference on Human-Robot Interaction (HRI '27), Industry White Papers*.

2025 **Lee, C. P.***, Choi, J.*, & Mutlu, B. [MAP: Multi-user Personalization with Collaborative LLM-powered Agents](#). *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*.

Wang, X. J., **Lee, C. P.**, & Mutlu, B. [LearnMate: Enhancing Online Education with LLM-Powered Personalized Learning Plans and Support](#). *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*.

2024 **Lee, C. P.** [Design, Development, and Deployment of Context-adaptive AI Systems for Enhanced User Adoption](#). *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*.

2023 **Lee, C. P.**, Cagiltay, B., Sullivan, D., & Mutlu, B. [Demonstrating the Potential of Interactive Product Packaging for Enriching Human-Robot Interaction](#). *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI '23)*.

PREPRINTS (1)

2026 **Lee, C. P.**, Wang, X. J., Albarghouthi, A., Porfirio, D., & Mutlu, B. [U-Define: Designing User Workflows for Hard and Soft Constraints in LLM-Based Planning](#). *arXiv preprint, arXiv:2605.02765*.

OTHER PUBLICATIONS (2)

2019 Nam, J., **Lee, C. P.**, Patankar, A. A., Wang, H., Li, Y., & Moon, H. Object Classification for Domestic Waste based on Convolutional Neural Networks. *Proceedings of the Korean Institute of Broadcast and Media Engineers Fall Conference*.

2018 Jung, J. Y., **Lee, C. P.**, Kim, J. H., Kim, S. I., Kim, J. I., Kong, H. S., Lee, Y. B., Lee, Y. L., & Kang, J. W. Increasing Video Compression Efficiency using Residual Data Distribution and Transform. *Proceedings of the WISET Junior Science & Technology Research Reports*, Vol. 1, pp. 350–354.

Talks

INVITED TALKS

2025 *Ploutos AI Community*, April 2025

“VeriPlan: Integrating Formal Verification and LLMs into End-User Planning”

Carnegie Mellon University, HRI Reading Group, March 2025

“Formal Methods for User-Centered AI: Enabling Effective Real-World Integration”

2024 *The University of Texas at Austin, School of Information*, hosted by Dr. Min Kyung Lee, September 2024

“Designing User-Centered Tools for Real-World AI Integration”

CONFERENCE PRESENTATIONS

2025 *ACM CHI Conference on Human Factors in Computing Systems (CHI 2025)*, Yokohama, Japan

“VeriPlan: Integrating Formal Verification and LLMs into End-User Planning”

- 2024 *ACM Designing Interactive Systems Conference (DIS 2024)*, Copenhagen, Denmark
 “The AI-DEC: A Card-based Design Method for User-centered AI Explanations”
- ACM Designing Interactive Systems Conference (DIS 2024)*, Copenhagen, Denmark
 “REX: Designing User-centered Repair and Explanations to Address Robot Failures”
- 2022 *ACM CHI Conference on Human Factors in Computing Systems (CHI 2022)*, New Orleans, LA
 “The Unboxing Experience: Exploration and Design of Initial Interactions Between Children and Social Robots”

DOCTORAL CONSORTIA

- 2026 *Future Leaders of AI Doctoral Consortium, ACM AI Leadership Summit 2026*, Atlanta, GA
 “Designing Responsible AI Workflows: Preserving Control and Supporting Alignment”
- 2024 *Doctoral Consortium, ACM CHI Conference on Human Factors in Computing Systems (CHI 2024)*, Honolulu, HI
 “Design, Development, and Deployment of Context-adaptive AI Systems for Enhanced User Adoption”

Teaching & Mentoring

TEACHING

- 2025 *Guest Lecture*, Theory and Practice in Computer Science Education, UW–Madison, November 2025
 “Critically Conscious Computing”
- Guest Lecture*, Autonomous Robotics, UW–Madison, April 2025
 “Introduction to Human-Robot Interaction (HRI)”
- 2024 *Guest Lecture*, Human-Artificial Intelligence Interaction, The University of Texas at Austin, September 2024
 Hands-on session with the AI-DEC design method
- 2020–2021 *Teaching Assistant*, CS 354: Machine Organization and Programming, UW–Madison, Fall 2020 and Spring 2021
Instructor: Dr. Debra Deppeler

STUDENT MENTORING

- 2024– *Xinyu Jessica Wang*, UW–Madison, now graduate student at UW–Madison
 Guided development for translation on natural language translation to formal language properties using LLMs
 Mentor for undergraduate senior thesis
- 2024–2025 *Kevin Zhao*, UW–Madison
 Guided development for model checking techniques using LLMs
 Mentor for undergraduate senior thesis
- 2022–2023 *Hongyu Fu*, UW–Madison, now graduate student at Carnegie Mellon University
 Guided development of A* algorithm for route planning in robot failures
- Colin Li*, UW–Madison, now graduate student at University of Southern California
 Guided algorithm development using LLMs for robot navigation

OUTREACH

- 2024 UW–Madison Grandparents University
 Served as one of four instructors to teach grandparents and grandchildren how to program robots
- 2023 High school outreach
 Hosted half-day lab visit for robot demonstration and discussion panel for computer science majors

Service

COMMUNITY

2025–2026 Webmaster, Human-Computer Interaction Consortium (HCIC), <https://hcic.org>

CONFERENCE REVIEWING

2022– ACM CHI Conference on Human Factors in Computing Systems (CHI)

2023– ACM/IEEE International Conference on Human-Robot Interaction (HRI)

2024– ACM Symposium on User Interface Software and Technology (UIST)

WORKSHOP REVIEWING

2025 IEEE International Conference on Robotics and Automation (ICRA 2025), Workshop on “Human-Centered Robot Learning in the Era of Big Data and Large Models”

Selected Press

2024 *Alhub*, [Bridging the Gap between User Expectations and AI Capabilities: Introducing the AI-DEC Design Tool](#), August 2024

References

Bilge Mutlu, Stephen Wright Professor, Department of Computer Sciences, University of Wisconsin–Madison
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Jodi Forlizzi, Herbert A. Simon Professor, Human-Computer Interaction Institute, Carnegie Mellon University
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Hae Won Park, Amazon Scholar, Amazon
robhwp@amazon.com